puzzle, one that, if realized, would be just about the best thing we could hope for. I'll describe it first in the most optimistic light, and then stress what still needs to be done.

If a wavefunction for an isolated electron shows that it has, say, a 50 percent chance of being here and a 50 percent chance of being there, we must interpret these probabilities using the full-fledged weirdness of quantum mechanics. Since both of the alternatives can reveal themselves by commingling and generating an interference pattern, we must think of them as equally real. In loose language, there's a sense in which the electron *is* at both locations. What happens now if we measure the electron's position with a nonisolated, everyday-sized laboratory instrument? Well, corresponding to the electron's ambiguous whereabouts, the pointer on the instrument has a 50 percent chance of pointing to this value and a 50 percent chance of pointing to that value. But because of decoherence, the pointer will *not* be in a ghostly mixture of pointing at both values; because of decoherence, we can interpret *these* probabilities in the usual, classical, everyday sense. Just as a coin has a 50 percent chance of landing heads and a 50 percent chance of landing tails, but lands *either* heads *or* tails, the pointer has a 50 percent chance of pointing to this value and a 50 percent chance of pointing to that value, but it will definitely point to one *or* the other.

Similar reasoning applies for all other complex, nonisolated objects. If a quantum calculation reveals that a cat, sitting in a closed box, has a 50 percent chance of being dead and a 50 percent chance of being alive—because there is a 50 percent chance that an electron will hit a booby-trap mechanism that subjects the cat to poison gas and a 50 percent chance that the electron misses the booby trap—decoherence suggests that the cat will *not* be in some absurd mixed state of being both dead and alive. Although decades of heated debate have been devoted to issues like What does it mean for a cat to be both dead and alive? How does the act of opening the box and observing the cat force it to choose a definite status, dead or alive?, decoherence suggests that long before you open the box, the environment has already completed billions of observations that, in almost no time at all, turned all mysterious quantum probabilities into their less mysterious classical counterparts. Long before you look at it, the environment has compelled the cat to take on one, single, definite condition. Decoherence forces much of the weirdness of quantum